Kai Fu

MPI MIS

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Curriculum Vitae

Employment

2025–present **Postdoc**, *Max Planck Institute for Mathematics in the Sciences (MPI-MiS), Leipzig*.

Education

2022–2025 **Ph.D. in Mathematics**, *IMB, Université de Bordeaux*.

2020–2022 **Master in Mathematics**, *Indiana University Bloomington*.

2017–2020 **Master in Mathematics**, *Nankai University*.

2013–2017 **Bachelor in Mathematics and Applied Mathematics**, *Beijing Institute of Technology*.

Research Interests

My research centers on surfaces (real two-dimensional manifolds), with an emphasis on the interplay between geometry, combinatorics, and dynamics. In particular, I study translation surfaces and, more broadly, flat cone surfaces, together with their moduli spaces.

Publications and Preprints

Point Processes and Gap Statistics for Translation Surfaces, Kai Fu, Anthony Sanchez, 2026, arXiv .

Siegel–Veech Measures of Convex Flat Cone Spheres, Kai Fu, *Compositio Mathematica*, 162(7) (2026), 1507–1573. arXiv, DOI .

Uniform Length Estimates for Trajectories on Flat Cone Surfaces, Kai Fu, To appear in *Annales de l'Institut Fourier*. arXiv .

Bounds on saddle connections for flat spheres, Kai Fu, Guillaume Tahar, *Journal of Topology*, 19(2) (2026). arXiv, DOI .

In Preparation

L^∞-Delaunay cells and linear subvarieties in the moduli space of Abelian differentials, *with Vincent Delecroix and Bradley Zykoski*.

Programming Project

I use computational methods in my research, often with **SageMath**. Currently, I am working with Vincent Delecroix on the computation of real linear subvarieties in the moduli spaces of Abelian differentials. More precisely, we develop algorithms to verify that a given germ defines a real linear subvariety. We also compute the boundary of a real linear subvariety in the multi-scale compactification. This project is being implemented in the package **veerer**.

Talks

Jul 15, 2026 **Geometry and Groups Seminar**, *Heidelberg University*, Heidelberg, Germany.

Jul 09, 2026 **Oberseminar Differentialgeometrie**, *Max Planck Institute for Mathematics*, Bonn, Germany.

Jun 02, 2026 **CMI-HIMR Summer School on Random Geometries and Random Matrices**, *University of Bristol*, (lightning talk), Bristol, United Kingdom.

Nov 25, 2025 **The floor to the junior researchers of Plato's tetrahedron's research network**, *Université de Montpellier*, Montpellier, France.

Nov 18, 2025 **UCL/KCL Junior Geometry Seminar**, *King's College*, London, United Kingdom.

Oct 08, 2025 **Géométrie et dynamique dans les espaces de modules**, *Institut Henri Poincaré (IHP)*, Paris, France.

Sep 25, 2025 **Fifth GALS meeting**, *Institut Henri Poincaré (IHP)*, Paris, France.

Aug 25, 2025 **Seminar**, *School of Mathematics, Sun Yat-sen University*, Guangzhou, China.

Aug 15, 2025 **Dynamical Systems Seminar**, *Shanghai Institute for Mathematics and Interdisciplinary Sciences*, Shanghai, China.

Jul 29, 2025 **Seminar**, *Chern Institute of Mathematics*, Tianjin, China.

Jul 17, 2025 **The 5th Korea-France Conference in Mathematics**, *Institut de Mathématique de Bordeaux*, Bordeaux, France.

Jun 27, 2025 **Séminaire de Géométrie**, *Institut de Mathématique de Bordeaux*, Bordeaux, France.

Nov 08, 2024 **Séminaire de Systèmes Dynamiques**, *Institut de mathématiques de Toulouse*, Toulouse, France.

Nov 07, 2024 **Groupe de travail**, *Institut de mathématiques de Toulouse*, Toulouse, France.

Oct 04, 2024 **Geometry Topoloy Grad and Postdoc Seminar**, *University of Toronto*, online.

Sep 10, 2024 **Riemann Surfaces: Random, Flat, and Hyperbolic Geometry**, *MFO*, (Lightning talk), Oberwolfach, Germany.

Oct 17, 2023 **Cours de Géométrie Algébrique et Géométrie Complexe de la fédération MARGAUx**, *Le Laboratoire de Mathématiques et Applications, Université de Poitiers Poitiers*, (Short talk), Poitiers, France.

Jun 08, 2023 **Student Seminar**, *YMSC, Tsinghua University*, Beijing, China.

Mar 16, 2023 **Reading group on mapping class groups**, *Institut de mathématiques de Bordeaux*, Bordeaux, France.

Visiting

Jul, 2026, MPIM Bonn (1 week)

Aug, 2025, School of Mathematics, Sun Yat-sen University (1 week)

Jul, 2025, Chern Institute of Mathematics (1 week)

Jun, 2023, YMSC, Tsinghua University (3 weeks)

Teaching Experience

Apr.–Jul. 2026 **Introduction to Translation Surfaces and Interval Exchange Transformations**, *Leipzig University*, Leipzig, Germany.
Co-instructor with Magali Jay; Master's-level course taught in English, two hours per week.

Organization

Oct–present, 2026 Co-organizer of the Geometry, Groups, and Dynamics Research Seminar, Leipzig

Mar, 2024 Co-organizer of the School on flat surfaces and interactions, Le Teich

Jan–May, 2023 Co-organizer of the Reading group on mapping class groups, Bordeaux

Conference Participation

Jun, 2026 Geometry, Dynamics, and Computer-Assisted Proofs, a conference in honor of Rich Schwartz, Heidelberg University, Heidelberg

Jun, 2026 CMI-HIMR Summer School on Random Geometries and Random Matrices, University of Bristol, Bristol

Sep, 2025 Fifth GALS meeting, IHP, Paris

Jul, 2025 The 5th Korea-France Conference in Mathematics, IMB, Bordeaux

Nov,2024 Riemann Surfaces: Random, Flat, and Hyperbolic Geometry, MFO, Oberwolfach

Jun, 2024 New Frontiers in Parabolic Dynamics and Renormalization, Department of Mathematics, University of Bologna, Bologna

Jun, 2024 Rencontre GALS, Institut Camille Jordan, Lyon

Mar, 2024 School on flat surfaces and interactions, Le Teich

Jan, 2024 Sage Days 125: combinatorics, number theory, dynamical systems and geometry, Le Teich

Jan, 2024 Research School - Hodege Theory and o-minimality, CIRM, Marseille

Dec,2023 Renormalization and Visualization in Geometry, Dynamics and Number Theory, CIRM, Marseille

Nov, 2023 The floor to the junior researcher of Plato's network Insitut Fourier, Grenoble

Oct, 2023 Cours de Géométrie Algébrique et Géométrie Complexe de la fédération MARGAUx, Université de Poitiers, Poitiers

Oct, 2023 Probability and Geometry in, on and of non-Euclidian spaces, CIRM, Marseille

Jul, 2023 Renormalization and Visualization for packing, billiard and surfaces, CIRM, Marseille

Feb, 2023	Sage Days 117: combinatorics, number theory, dynamical systems and geometry, Le Teich
Sep, 2022	Combinatorics, Dynamics and Geometry on Moduli Spaces, CIRM, Marseille
May, 2022	Structures on Surfaces, CIRM, Marseille
Apr, 2022	Mini-rencontre ANR MoDiff, LaBRI, Université de Bordeaux, Bordeaux
Jun, 2021	CMI-HIMR Dynamics and Geometry Online Summer School, University of Bristol

Languages

Chinese	native
English	fluent
French	basic